7. Assertion : Permanent hardness of water is removed by treatment with washing soda.

Reason : Washing soda reacts with soluble calcium and magnesium chlorides and sulphates in hard water to form insoluble carbonates

a) Both assertion and reason are true and reason is the correct explanation of assertion.

b) Both assertion and reason are true but reason is not the correct explanation of assertion.

c) Assertion is true but reason is false

d) Both assertion and reason are false

8. If a body of a fish contains 1.2 g hydrogen in its total body mass, if all the hydrogen is replaced with deuterium then the increase in body weight of the fish will be

a) 1.2 g b) 2.4 g c) 3.6 g d) $\sqrt{4.8}$ g

9. The hardness of water can be determined by volumetrically using the reagent

a) sodium thio sulphate b) potassium permanganate

c) hydrogen peroxide d) EDTA

10. The cause of permanent hardness of water is due to

a) $Ca(HCO_3)_2$ b) $Mg(HCO_3)_2$ c) $CaCl_2$ d) $MgCO_3$

11. Zeolite used to soften hardness of water is, hydrated

a) Sodium aluminium silicate b) Calcium aluminium silicate

c) Zinc aluminium borate d) Lithium aluminium hydride

12. A commercial sample of hydrogen peroxide marked as 100 volume H_2O_2, it means that

a) 1 ml of H_2O_2 will give 100 ml O_2 at STP

b) 1 L of H_2O_2 will give 100 ml O_2 at STP

c) 1 L of H_2O_2 will give 22.4 L O_2

d) 1 ml of H_2O_2 will give 1 mole of O_2 at STP

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